Module 6a slides: Multiple Oral Doses (worked examples)

Module 6 - Multiple Dosings · Lecture slides, worked examples · 28 Sep 2026

Name _____ Date _____

Example 1 An adult male (75 kg) received 250 mg tetracycline hydrochloride orally every 8 hours for 2 weeks. Tetracycline hydrochloride is about 75% bioavailable and has a V_D of 1.5 L/kg. The t½ is about 10 hours and the k_a is 0.9 hr^{-1}. When does the peak of the first dose occur?

Example 1 For the tetracycline regimen (250 mg orally, F 0.75, V_D 112.5 L, k 0.0693 hr^{-1}, k_a 0.9 hr^{-1}), the first dose peaks at 3.09 hours. What is C_{max} of the first dose?

Example 1 For the tetracycline regimen (250 mg orally every 8 hours, k 0.0693 hr^{-1}, k_a 0.9 hr^{-1}), when does the peak occur at steady state?

Example 1 For the tetracycline regimen (250 mg orally every 8 hours, F 0.75, V_D 112.5 L, k 0.0693 hr^{-1}), the steady-state peak occurs at 2.06 hours. What is C_{max} at steady state?

Example 1 For the tetracycline regimen (250 mg orally every 8 hours, F 0.75, V_D 112.5 L, k 0.0693 hr^{-1}, k_a 0.9 hr^{-1}), what is C_{min} at steady state?

Example 1 For the tetracycline regimen (250 mg orally every 8 hours, F 0.75, V_D 112.5 L, k 0.0693 hr^{-1}), what is Cavg at steady state?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

Example 1 Her key

Answer: 3.09 hr

WHAT THE STEM GIVES

weight	75 kg	*not needed: t_{max} has no volume in it*
D_0	250 mg	*not needed: t_{max} does not depend on the dose*
τ	every 8 hours	*not needed: the first dose has no accumulation*
duration	for 2 weeks	*not needed*
F	75% bioavailable	*not needed: F scales the height of the curve, not its timing*
V_D	1.5 L/kg	*not needed: t_{max} has no volume in it*
t½	about 10 hours	gives $k = \frac{0.693}{10} = 0.0693\ hr^{-1}$
k_a	0.9 hr^{-1}	the numerator of $\frac{k_a}{k}$ and of $k_a - k$

THE WORKING, ONE LINE AT A TIME

UNIT

$$k = \frac{0.693}{10\ hr} = 0.0693\ hr^{-1};\ V_D = (1.5\ L/kg)(75\ kg) = 112.5\ L$$

The half-life becomes k, and the per-kilogram volume is scaled to the patient. t_{max} needs only k and k_a; V_D is needed for the next part.

SETUP

$$t_{max} = \frac{\ln(k_a/k)}{k_a - k}$$

For the first dose the body starts empty, so the single-dose time to peak applies, with no τ in it. The dosing interval enters only at steady state.

ALGEBRA

$$t_{max} = \frac{\ln(0.9/0.0693)}{0.9 - 0.0693} = \frac{\ln(12.99)}{0.8307} = \frac{2.564}{0.8307} = 3.09\ hr$$

Reciprocal hours in the denominator leave hours. The logarithm has no units, because $\frac{k_a}{k}$ is a ratio of two rate constants in the same unit.

ROUND

$$t_{max} = 3.1\ hr$$

Her value: "something like 3-ish hours for T-Max", written as 3.1 hr in the working. The first-dose C_{max} is evaluated at this time in the next part.

> **DOES THE NUMBER MAKE SENSE?**

Absorption at $k_a = 0.9\ hr^{-1}$ has a half-life of $\frac{0.693}{0.9}$, under 1 hour, and elimination is 13 times slower, so the peak comes a few absorption half-lives in: about 3 hours, well inside the 8-hour interval.

Example 1 Her key

Answer: 1.35 mg/L

WHAT THE STEM GIVES

D_0	250 mg	in the numerator $F\ k_a\ D_0$
F	0.75	in the numerator $F\ k_a\ D_0$
V_D	112.5 L	in the denominator $V_D(k_a - k)$
k	0.0693 hr^{-1}	in $k_a - k$ and in $e^{-k\ t_{max}}$
k_a	0.9 hr^{-1}	in the numerator, in $k_a - k$ and in $e^{-k_a\ t_{max}}$
t_{max}	3.09 hours	t in both exponentials

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{max} = \frac{F\ k_a\ D_0}{V_D(k_a - k)}\ (e^{-k\ t_{max}} - e^{-k_a\ t_{max}})$$

This is the single-oral-dose equation with t set to t_{max}. The interval does not enter, because this is the first dose.

ALGEBRA

$$\frac{F\ k_a\ D_0}{V_D(k_a - k)} = \frac{(0.75)(0.9\ hr^{-1})(250\ mg)}{(112.5\ L)(0.9 - 0.0693\ hr^{-1})} = \frac{168.75}{93.45} = 1.806\ mg/L$$

mg times hr^{-1} over L times hr^{-1} leaves mg/L. This prefactor is not the peak: it multiplies a bracket that is always less than 1, so the peak is smaller than 1.806 mg/L.

ALGEBRA

$$e^{-0.0693 \times 3.09} - e^{-0.9 \times 3.09} = 0.807 - 0.062 = 0.745$$

At the peak most of the dose has been absorbed (the k_a term, 0.062, is nearly gone) and little has been eliminated (the k term is still 0.807), so the bracket is close to 0.75.

ALGEBRA

$$C_{max} = 1.806 \times 0.745 = 1.35\ mg/L$$

The prefactor times the bracket: $1.806 \times 0.745 = 1.35$. Both factors are known only once t_{max} has been found, which is why t_{max} came first.

ROUND

$$C_{max} = 1.35\ mg/L$$

Her value: "1.35 mg per liter for C-Max". The steady-state peak of 3.4 mg/L is compared with this first-dose peak.

DOES THE NUMBER MAKE SENSE?

If the whole absorbed dose were in the body at once the level would be $\frac{0.75 \times 250 \text{ mg}}{112.5 \text{ L}}$, a little under 2 mg/L; by 3.09 hours some has been eliminated and a little is still unabsorbed, so C_{max} is below that: 1.35 mg/L.

Example 1 Her key

Answer: 2.06 hr

WHAT THE STEM GIVES

D_0	250 mg	*not needed: $t_{max}\infty$ has no dose in it*
τ	every 8 hours	in both $1 - e^{-k\tau}$ and $1 - e^{-k_a\tau}$
k	0.0693 hr^{-1}	in $k_a - k$, in $e^{-k\tau}$ and in the denominator $k(1 - e^{-k_a\tau})$
k_a	0.9 hr^{-1}	in $k_a - k$, in $e^{-k_a\tau}$ and in the numerator

THE WORKING, ONE LINE AT A TIME

SETUP

$$t_{max}\infty = \frac{1}{k_a - k} \ln\left[\frac{k_a(1 - e^{-k\tau})}{k(1 - e^{-k_a \tau})}\right]$$

At steady state the interval enters the time to peak. The two bracketed terms carry τ with k and with k_a; they are what make this equation differ from the single-dose t_{max}.

ALGEBRA

$$1 - e^{-0.0693 \times 8} = 1 - 0.574 = 0.426; \; 1 - e^{-0.9 \times 8} = 1 - 0.0007 = 0.999$$

Over one 8-hour interval 43 per cent of the drug is eliminated, and absorption is complete (0.999) because k_a is large.

ALGEBRA

$$\ln\left[\frac{0.9 \times 0.426}{0.0693 \times 0.999}\right] = \ln\left(\frac{0.383}{0.0692}\right) = \ln(5.53) = 1.710$$

The ratio inside the logarithm is smaller than the single-dose $\frac{k_a}{k}$ of 12.99, which is why the steady-state peak comes earlier.

ALGEBRA

$$t_{max}\infty = \frac{1.710}{0.8307 \text{ hr}^{-1}} = 2.06 \text{ hr}$$

Dividing by $k_a - k = 0.8307 \text{ hr}^{-1}$ leaves hours, because the logarithm has no units. $\frac{1.710}{0.8307} =$ 2.06 hr, about an hour earlier than the first-dose peak.

ROUND

$$t_{max}\infty = 2.06 \text{ hr}$$

Her value: "T-Max at steady state. I got 2.06 hours." It is shorter than the first-dose 3.1 hours, and it is the time at which C_{max} at steady state is evaluated.

DOES THE NUMBER MAKE SENSE?

At steady state drug left from earlier doses is eliminated while the new dose absorbs, so the peak arrives earlier than the first-dose peak and well inside the 8-hour interval: 2.06 hours, between 0 and 8.

Example 1 Worked by the drill

Answer: 3.39 mg/L

WHAT THE STEM GIVES

D_0	250 mg	in the numerator FD_0
τ	every 8 hours	in the accumulation factor $1 - e^{-k\tau}$
F	0.75	in the numerator FD_0
V_D	112.5 L	the denominator of $\frac{FD_0}{V_D}$ = 1.667 mg/L
k	0.0693 hr^{-1}	in $e^{-k\tau}$ and in $e^{-k\, t_{max}^{\infty}}$
t_{max}^{∞}	2.06 hours	t in $e^{-k\, t_{max}^{\infty}}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{max}^{\infty} = \frac{FD_0}{V_D}\left(\frac{1}{1 - e^{-k\tau}}\right) e^{-k\, t_{max}^{\infty}}$$

The steady-state peak for an oral dose: the absorbed dose per litre, raised by the accumulation factor, then declined for the time to peak.

ALGEBRA

$$\frac{FD_0}{V_D} = \frac{(0.75)(250\text{ mg})}{112.5\text{ L}} = 1.667\text{ mg/L}$$

The concentration the absorbed dose would give if it were all in the body at once: 75 per cent of 250 mg, spread through 112.5 L.

ALGEBRA

$$\frac{1}{1 - e^{-(0.0693)(8)}} = \frac{1}{1 - 0.574} = \frac{1}{0.426} = 2.35$$

The accumulation factor for an 8-hour interval and a 10-hour half-life. It is more than double, because less than one half-life passes between doses.

ALGEBRA

$e^{-0.0693 \times 2.06} = e^{-0.143} = 0.867$; $C_{max}^{\infty} = 1.667 \times 2.35 \times 0.867 = 3.40$ mg/L

The decline over the 2.06 hours to the peak, from her t_{max} at steady state. The product of the three factors, 1.667 × 2.35 × 0.867, is the peak at steady state.

ROUND

$C_{max}^{\infty} = 3.4$ mg/L

Her spoken value is "Steady state, 3.3". Carried unrounded with t_{max}^{∞} = 2.06 the product is 3.395 mg/L, and with 2.1, as written in the working, 3.386. It is higher than the first-dose 1.35 mg/L, as accumulation predicts.

DOES THE NUMBER MAKE SENSE?

One dose alone gives $\frac{FD_0}{V_D}$ = 1.667 mg/L and accumulation multiplies it by 2.35; the 2.06 hours of elimination before the peak leave 0.867 of that, so C_{max}^{∞} = 1.667 × 2.35 × 0.867 = 3.40 mg/L, above 1.667.

Example 1 Her key

Answer: 2.44 mg/L

WHAT THE STEM GIVES

D_0	250 mg	in the numerator k_a F D_0
τ	every 8 hours	in $e^{-k\tau}$ twice: accumulation and the decay to the trough
F	0.75	in the numerator k_a F D_0
V_D	112.5 L	in the denominator $V_D(k_a - k)$
k	0.0693 hr^{-1}	in $k_a - k$ and in $e^{-k\tau} = 0.574$
k_a	0.9 hr^{-1}	in the numerator and in $k_a - k = 0.8307$

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{min}^{\infty} = \frac{k_a F D_0}{V_D(k_a - k)} \left(\frac{1}{1 - e^{-k\tau}}\right) e^{-k\tau}$$

The steady-state trough for an oral dose: the single-dose prefactor, raised by the accumulation factor, declined for one full interval.

ALGEBRA

$$\frac{k_a F D_0}{V_D(k_a - k)} = \frac{(0.9\ hr^{-1})(0.75)(250\ mg)}{(112.5\ L)(0.8307\ hr^{-1})} = 1.806\ mg/L$$

The same prefactor as in the first-dose C_{max} calculation, since nothing in it changes with repeated dosing: F, k_a, V_D and k are set by the drug, the product and the patient, and D_0 is the same dose each time.

ALGEBRA

$$\frac{1}{1 - e^{-(0.0693)(8)}} = 2.35;\ e^{-(0.0693)(8)} = 0.574$$

The accumulation factor for τ = 8 hr and k = 0.0693 hr^{-1}, and the fraction of a dose remaining after one full interval, $e^{-0.5544} = 0.574$.

ALGEBRA

$$C_{min}^{\infty} = 1.806 \times 2.35 \times 0.574 = 2.44\ mg/L$$

The product of the three factors: 1.806 × 2.35 × 0.574 = 2.44 mg/L. The trough is lower than the peak by the decline over the rest of the interval after the peak.

ROUND

$$C_{min}^{\infty} = 2.44\ mg/L$$

Her value: "I'm getting 2.4-ish." The steady-state trough, 2.44 mg/L, is higher than the first-dose peak of 1.35 mg/L.

DOES THE NUMBER MAKE SENSE?

The coefficient 1.806 mg/L times accumulation 2.35 is the level a bolus-like dose would reach; one interval of decline leaves 0.574 of it: 2.44 mg/L, more than half of 1.806 × 2.35 and above zero.

Example 1 Worked by the drill

Answer: 3.01 mg/L

WHAT THE STEM GIVES

D_0	250 mg	in the numerator FD_0 = 187.5 mg
τ	every 8 hours	in the denominator V_D k τ = 62.37 L
F	0.75	in the numerator FD_0
V_D	112.5 L	in the denominator V_D k τ
k	0.0693 hr^{-1}	in the denominator V_D k τ

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{avg}^{\infty} = \frac{FD_0}{V_D\, k\, \tau}$$

The same average as for repeated bolus doses, with F less than 1 because only 75 per cent of each oral dose is absorbed. Cl_T (total clearance) = V_D k.

ALGEBRA

$FD_0 = (0.75)(250\text{ mg}) = 187.5\text{ mg}$; $V_D\, k\, \tau = (112.5\text{ L})(0.0693\text{ hr}^{-1})(8\text{ hr}) = 62.37\text{ L}$

L times hr^{-1} times hr leaves litres, so the result is mg over L. V_D k is the clearance, 7.80 L/hr, and clearance times τ is 62.37 L.

ALGEBRA

$$C_{avg}^{\infty} = \frac{(0.75)(250\text{ mg})}{(112.5\text{ L})(0.0693\text{ hr}^{-1})(8\text{ hr})} = \frac{187.5\text{ mg}}{62.37\text{ L}} = 3.01\text{ mg/L}$$

The average over one interval at steady state: 187.5 mg absorbed per interval, divided by the 62.37 L of plasma cleared per interval.

ROUND

$$C_{avg}^{\infty} = 3.01\text{ mg/L}$$

Computed from her inputs. The average, 3.01 mg/L, lies between the trough, 2.44, and the peak, 3.39 mg/L, and above their midpoint, 2.915.

DOES THE NUMBER MAKE SENSE?

Over one 8-hour interval 62.37 L of plasma is cleared, and the 187.5 mg absorbed per interval spread through it is $\frac{187.5}{62.37}$ = 3.01 mg/L, above zero and between the steady-state trough and peak.
